# Solar Power Prediction

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***Abstract*—This project develops a comprehensive machine learning pipeline to forecast AC power output for a solar photovoltaic (PV) plant using high-resolution operational and weather sensor data. The dataset, collected over 34 days at 15-minute intervals, integrates inverter-level generation metrics with plant-level environmental measurements. The study conducts extensive exploratory data analysis (EDA), feature engineering, and model benchmarking across Linear Regression, Random Forest, XGBoost, and Deep Neural Networks.**

**The results reveal that irradiation is the dominant driver of AC power generation, with module temperature and ambient temperature contributing secondary but meaningful effects. Ensemble models—particularly XGBoost—demonstrated superior predictive performance, achieving an $R^2$ of 0.9865 and an RMSE of 45.63 kW. These findings highlight the potential of machine learning to enhance solar plant operations, support grid stability, and enable data-driven energy planning.**

***Index Terms*—Solar Forecasting, Machine Learning, XGBoost, Random Forest, Renewable Energy, Predictive Modeling, Data Science.**

## I. Introduction

Solar energy is a cornerstone of global renewable energy strategies, offering a clean and sustainable alternative to fossil fuels. However, solar power generation is inherently intermittent due to its dependence on weather conditions such as irradiation, temperature, and cloud cover. This variability introduces operational challenges for grid operators, who must continuously balance electricity supply and demand.

Accurate short-term forecasting of solar power generation is therefore essential for improving grid stability, optimizing inverter scheduling, enhancing economic planning, and predicting equipment stress periods. Traditional forecasting approaches rely on physical models or numerical weather prediction (NWP), but these methods often struggle with short-term fluctuations. Machine learning offers a powerful alternative by learning nonlinear relationships directly from historical data.

This project aims to build a robust predictive model capable of estimating AC power output using real-world sensor data. The work includes EDA, feature engineering, model comparison, hyperparameter tuning, and deployment preparation.

## II. Related Work

Solar power forecasting has become a critical research area as global energy systems increasingly depend on renewable sources. The inherent variability of solar radiation—driven by atmospheric conditions such as cloud cover, temperature, and humidity—makes accurate forecasting essential for grid reliability and operational planning. Over the past decade, researchers have explored a wide range of statistical, machine learning, and deep learning approaches to improve prediction accuracy.

### A. Traditional and Statistical Forecasting Approaches

Early solar forecasting methods relied heavily on physical models and numerical weather prediction (NWP). These models simulate atmospheric processes to estimate future irradiation levels, but they often struggle with short-term fluctuations and require significant computational resources. Statistical models such as ARIMA and persistence forecasting provide simple baselines but fail to capture nonlinear interactions inherent in PV systems [1].

### B. Machine Learning for Solar Forecasting

Machine learning approaches have gained prominence due to their ability to learn complex, nonlinear patterns directly from data. Tree-based ensemble models—such as Random Forests and Gradient Boosting—consistently outperform linear models in structured solar datasets [2]. XGBoost, in particular, has demonstrated strong performance due to its regularization capabilities and robustness to overfitting. Li et al. showed that XGBoost outperformed Support Vector Regression (SVR) and Multi-Layer Perceptrons (MLPs) in short-term PV forecasting tasks [3].

### C. Deep Learning and Temporal Modeling

Deep learning has also been widely explored for solar forecasting. Feed-forward neural networks (FNNs) perform well on high-resolution sensor data, while recurrent architectures such as LLSTMs and GRUs excel at capturing temporal dependencies in solar radiation patterns [4]. However, deep models require large datasets and careful hyperparameter tuning, making them less practical for smaller datasets like the one used in this project.

### D. Feature Importance and Environmental Drivers

Across the literature, irradiation consistently emerges as the most influential predictor of solar power output. Studies confirm that module temperature, ambient temperature, and humidity also affect PV efficiency, but their impact is secondary compared to solar radiation intensity [1], [5]. Time-based features such as hour-of-day and day-of-year help capture diurnal and seasonal patterns, improving model stability and interpretability.

### E. Gaps Addressed by This Project

This project contributes to the field by:

- Using real-world, high-frequency (15-minute) operational data
- Comparing multiple model families under identical conditions
- Conducting feature importance analysis to validate physical relationships
- Preparing a deployment-ready model suitable for real-time applications

## III. Methodology

This study follows a structured workflow consisting of data acquisition, preprocessing, exploratory analysis, feature engineering, model development, and evaluation.

### A. Data Acquisition

Two datasets were used, collected from a solar power plant in India over 34 days at 15-minute intervals:

*1) Generation Data (68,778 records):* Includes inverter-level metrics:

- DC_POWER
- AC_POWER (target variable)
- DAILY_YIELD
- TOTAL_YIELD
- INVERTER_ID
- DATE_TIME

*2) Weather Sensor Data (3,182 records):* Includes plant-level environmental measurements:

- AMBIENT_TEMPERATURE
- MODULE_TEMPERATURE
- IRRADIATION
- SENSOR_ID
- DATE_TIME

The datasets were merged on DATE_TIME, resulting in a unified dataset of 68,774 records.

### B. Data Preprocessing

Key preprocessing steps included:

- Datetime parsing and extraction of hour, minute, month, and day-of-year
- Handling missing values (none remained after merging)
- Scaling using MinMaxScaler for neural network models
- Train-test split (80% training, 20% testing)

### C. Exploratory Data Analysis

EDA revealed strong and intuitive relationships:

*1) Irradiation vs AC Power:* A clear positive correlation was observed, confirming irradiation as the primary driver of AC power.

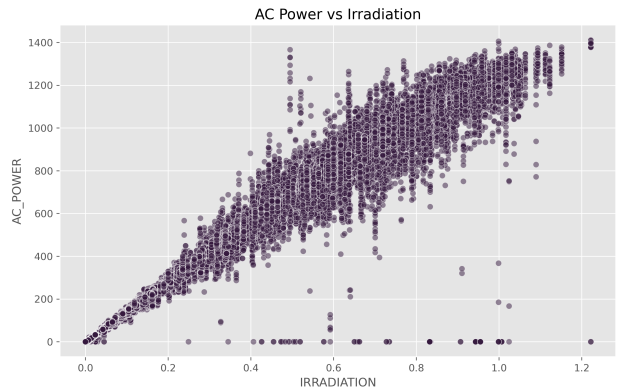


Fig. 1. AC Power vs Irradiation Scatter Plot

*2) Correlation Heatmap:* Irradiation, module temperature, and DC power showed the strongest correlations with AC power.

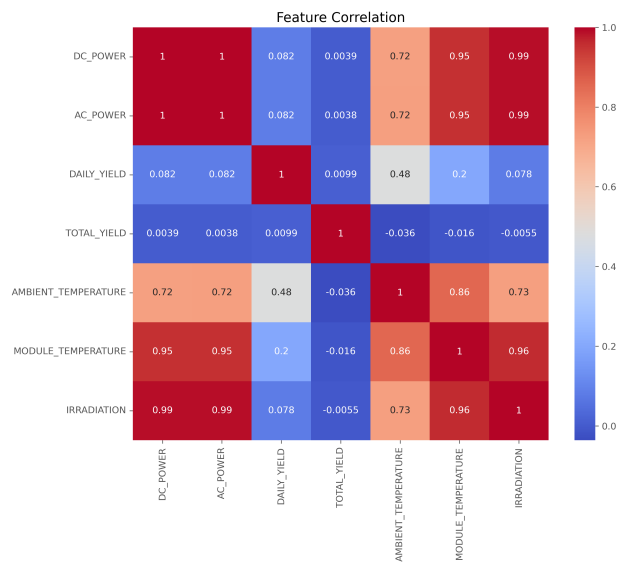


Fig. 2. Feature Correlation heatmap

*3) Hourly Boxplots:* AC power peaks between 10 AM and 2 PM, reflecting typical solar production cycles.

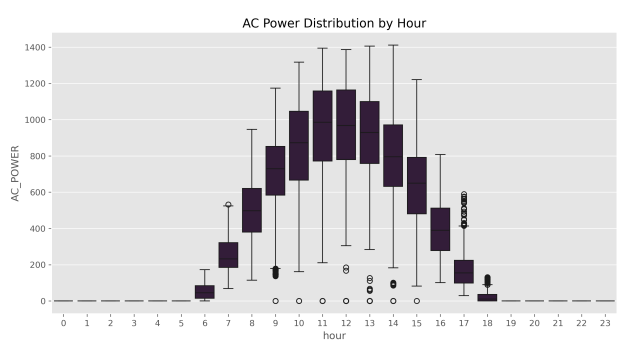


Fig. 3. AC power distribution by hour

*4) Pairplots:* Nonlinear relationships between temperature, irradiation, and power were evident.

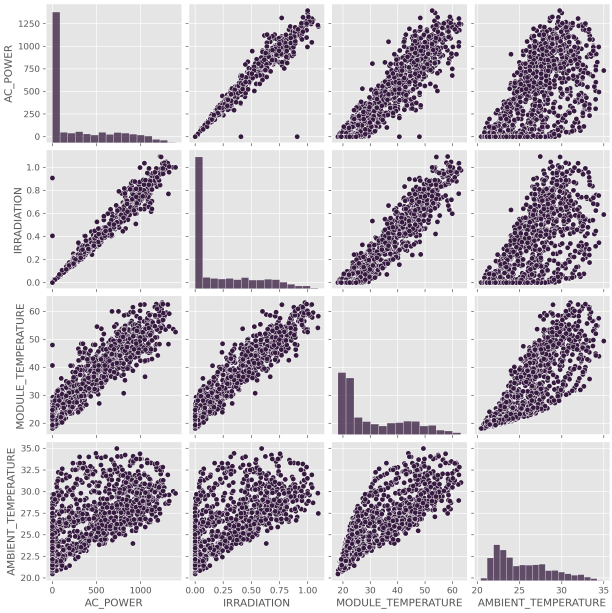


Fig. 4. Pairplot of key metrics

### D. Feature Engineering

Selected features:

- AMBIENT_TEMPERATURE
- MODULE_TEMPERATURE
- IRRADIATION
- hour
- minute

### E. Machine Learning Models

Four categories of models were developed:

1) Linear Regression (Baseline)
2) Random Forest Regressor
3) XGBoost Regressor
4) Deep Neural Networks (Keras/TensorFlow)

Hyperparameter tuning was performed for Random Forest and XGBoost using GridSearchCV.

## IV. Results

### A. Model Comparison

TABLE I
MODEL PERFORMANCE COMPARISON

| Model | RMSE (kW) | $R^2$ |
|---|---|---|
| XGBoost (Tuned) | **45.63** | **0.9865** |
| Random Forest (Tuned) | 45.64 | 0.9865 |
| Random Forest (Base) | 45.67 | 0.9865 |
| XGBoost (Base) | 45.97 | 0.9863 |
| Linear Regression | 55.36 | 0.9801 |
| DNN (Base) | 51.14 | 0.9831 |
| DNN (Deep) | 64.44 | 0.9731 |

### B. Final Evaluation of Winning Model

The tuned XGBoost Regressor was selected as the best model:

- RMSE: 45.63 kW
- $R^2$: 0.9865

*(Figure: XGBoost Predicted vs Actual Scatter Plot)*

### C. Feature Importance Analysis

XGBoost and Random Forest both identified IRRADIATION as the overwhelmingly dominant feature, followed by:

1) Irradiation
2) Module Temperature
3) Ambient Temperature
4) Hour
5) Minute

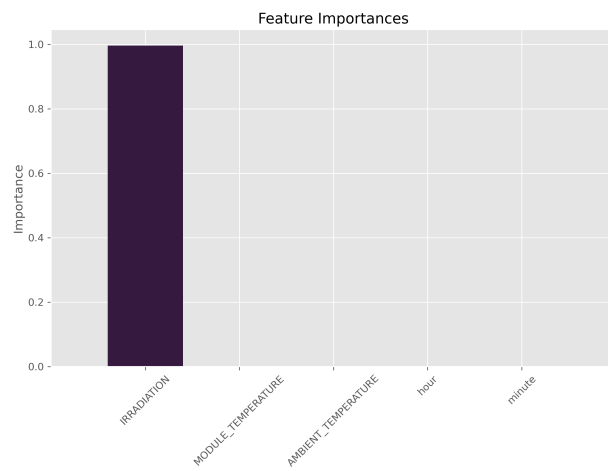


Fig. 5. Feature Importance Bar Chart

## V. Model Deployment

To support real-world use, the following components were saved:

- `best_model.pkl` — Tuned XGBoost model
- `scaler.pkl` — MinMaxScaler for preprocessing
- Feature list — Ensures consistent input structure

These components enable seamless integration into dashboards, APIs, or automated forecasting systems. For this project, a web-based application was developed using Docker and Flask.

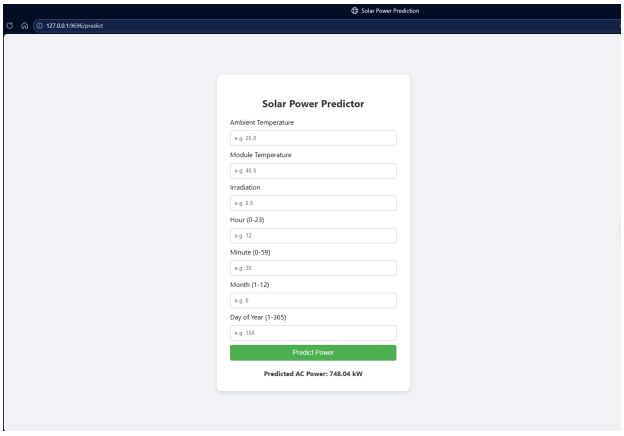


Fig. 6. Power Prediction web-application

## VI. Discussion

### A. Importance of Irradiation

The dominance of irradiation aligns with physical principles of photovoltaic generation. Temperature variables contribute but play a secondary role.

### B. Model Performance Insights

Tree-based ensemble models outperform linear and neural network models due to their ability to capture nonlinear interactions and handle skewed distributions.

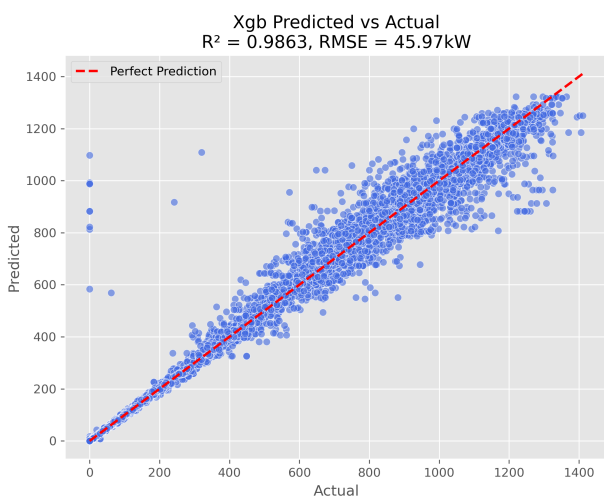


Fig. 7. Xgb predicted vs actual values

### C. Limitations

- Dataset covers only 34 days, limiting seasonal generalization
- Weather data is plant-level, while generation data is inverter-level
- No cloud-cover or wind-speed data included

### D. Future Work

- Incorporate satellite-based cloud cover
- Use LSTM networks for time-series forecasting
- Expand dataset to multiple seasons

## VII. Conclusion

This project successfully developed a high-accuracy machine learning pipeline for forecasting solar AC power output. Through extensive EDA, feature engineering, and model comparison, the tuned XGBoost model emerged as the best performer, achieving an $R^2$ of 0.9865. The findings reinforce the critical role of irradiation in solar power generation and demonstrate the value of data-driven forecasting for improving grid stability, operational planning, and renewable energy integration.

## References

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## Self-Declaration

**Beven Mpofu**

Led the overall project design and implementation. Conducted data cleaning, preprocessing, and exploratory data analysis. Developed baseline regression models and implemented advanced models including Random Forest, XGBoost, and Deep Neural Networks. Performed hyperparameter tuning, model evaluation, and interpretability analysis. Prepared the full written report and visualizations.

**Sibonginkosi T. Nkashe**

Assisted with data preprocessing, feature engineering, and correlation analysis. Contributed to the development of the machine learning pipeline and supported model training and validation. Helped prepare figures, tables, and documentation.

**Memory Mhembere**

Focused on literature review, dataset documentation, and background research on solar power forecasting. Contributed to writing the introduction, related work, and problem formulation sections. Assisted with preparing the final presentation and proofreading the report.

**Prince Tawiah**

Implemented additional experiments, including ensemble comparisons and error analysis. Contributed to SHAP-based interpretability and model comparison. Assisted with compiling results and formatting the final report.

## Acknowledgment

We acknowledge the open-source datasets and Python libraries that supported this analysis, as well as the guidance from our Professor Dr. Sujan Kumar Roy.